

Question

Compute the time complexity using frequency count for the following:

1)

for (i = 1; i <= 10; i = i + 2) // steps: 1+1+2 freq: 1 + 11 + 20 = 32
{
 i = i & 10; // steps: 1+1 freq: 10 x 2 = 20
 printf("%d", i); // steps: 1 freq: 10 = 10
}

Total = 32 + 20 + 10 = 62

2)

i = 1; // steps: 1 freq: 1
while (i <= 50); // steps: 1 freq: 1
{
 printf("hello");
}

Total = 1 + 1 = 2

3)

for (i = 1; i <= 20; i++); // steps: 1+1+1 freq: 3

Total = 3

4)

for (i = 1; i <= n; i = i++) // steps: 1+1+2 freq: 1 + (n+1) + 2n = 3n + 2
{
 printf("hello"); // steps: 1 freq: n
 printf("world"); // steps: 1 freq: n
 printf("i = %d", i); // steps: 1 freq: n
}

Total = (3n + 2) + n + n + n = 6n + 2